

Temperature trends in Geneva

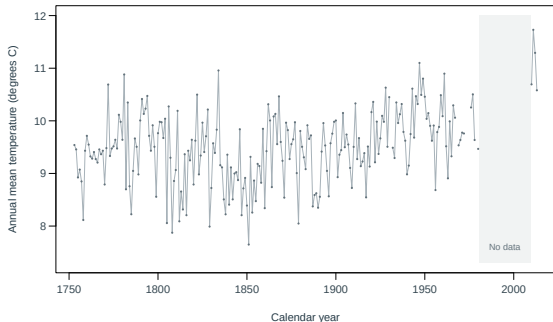
Does one constant trend describe the record?
Linear diagnostics, structural change and bootstrap inference

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Station08 1753–2013 226 observations

A long record with substantial missing years



OBSERVED YEARS

226

1753–2013

MISSING YEARS

35

Across five gaps

LONGEST GAP

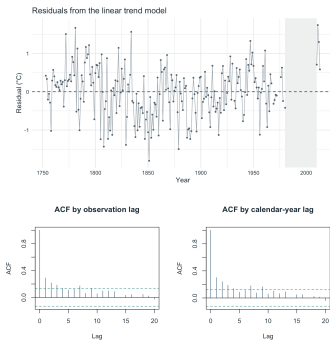
29 years

1981–2009

All supplied observations retained; no interpolation.

Geneva, Switzerland; GHCN-M v4 QCF adjusted series. Annual means require at least ten valid months. Lines stop across gaps.

The linear trend leaves dependent residuals



$$x_i = (\text{Year}_i - 1850)/10$$

$$\hat{y}_i = 9.4481 + 0.0337x_i$$

Slope: °C per decade.

Intercept: fitted temperature in 1850.

Diagnostic	Statistic	p
DW	1.4147	0.0002
BP	0.4173	0.5183

DW rejection region

$d < 1.7549$ or $d > 2.2695$.

BP does not reject a linear variance trend.

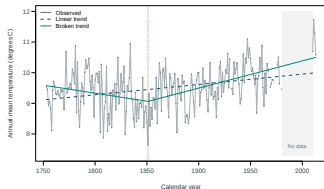
Residual dependence motivates the dependent wild bootstrap.

DW: 9,999 iid Gaussian-null draws, separate two-sided cutoffs. BP: nR^2 from e_i^2 on x_i , nominal χ_1^2 reference. ACF bands are descriptive.

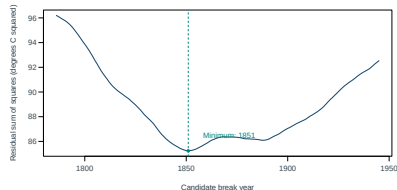
A continuous slope change fits best in 1851

$y_i = \alpha + \beta x_i + \delta(x_i - x_k)_+ + \varepsilon_i$: a slope change with no level jump.

Fitted trends



Complete date search



BEFORE THE BREAK

-0.0525

°C per decade

CHANGE IN SLOPE

+0.1412

°C per decade

AFTER THE BREAK

+0.0887

°C per decade

A better fit alone does not establish a structural break.

OLS at each date; 15% observation-count trimming, candidates 1786–1945. $H_0 : \delta = 0$;
 $F_T = RSS_0 - \min_k RSS_k = 12.2106 \text{ } ^\circ\text{C}^2$. RSS reduction: 12.53%.

Bootstrap choices follow the diagnostics

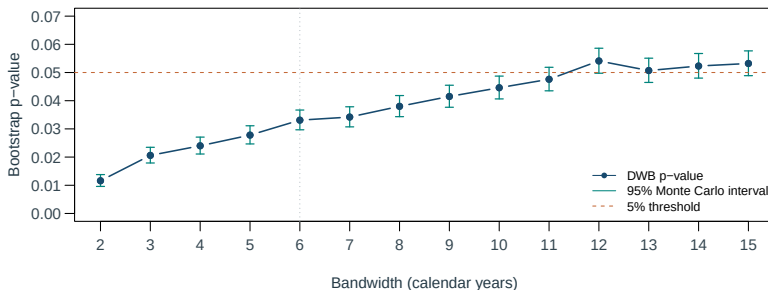
Method	Error draw	Preserves
IID residual	Resample residuals	Common distribution
Independent wild	Residual $\times$ independent sign	Local residual scale
Dependent wild	Residual $\times$ correlated weight	Scale; approximate dependence

$$\text{Corr}(w_i, w_j) = \exp(-|s_i - s_j|/\ell), \quad \ell = 6 \text{ years}$$

- Generate under the fitted linear null using its residuals.
- Repeat the full date search in every bootstrap sample.

9,999 draws per test. Primary DWB: $\ell = \text{round}(226^{1/3}) = 6$ calendar years, a heuristic. Sensitivity grid: 2–15 years.

Break evidence changes with the bandwidth

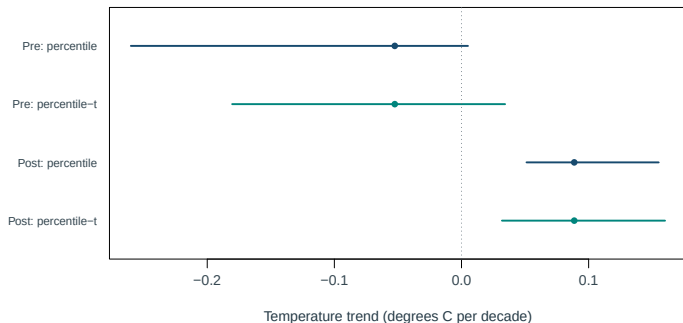


IID / independent wild: $p = 0.0001$; primary DWB ($\ell = 6$): $p = 0.0331$.

At 5%: bandwidths 2–11 reject; 12–15 do not.

9,999 draws per bandwidth, full break search in each draw. Dashed line: 5%; dotted line: primary $\ell = 6$. Bars: 95% Wilson intervals for Monte Carlo tail probabilities at fixed bandwidths.

The later fitted slope is positive



Pre-break

Both intervals include zero.

Post-break

Both intervals exclude zero.

Generate under the fitted broken alternative; reselect the date in all 4,999 draws.

DWB bandwidth six. Percentile-t intervals use calendar-time HAC studentisation. Both interval types are conditional on the broken-trend specification; they do not independently test for a break.

Eight scenarios assess size and power

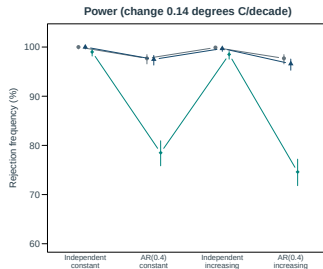
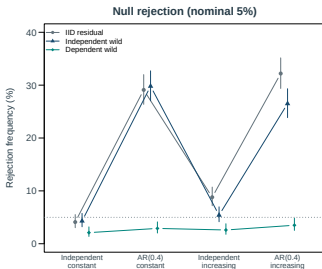
$$y_t = 9.5 - 0.05x_t + \delta(x_t - x_{113})_+ + \varepsilon_t, \quad \varepsilon_t = \rho\varepsilon_{t-1} + \sigma_t u_t$$

Factor	Values
Slope change δ	0 or 0.14 °C per decade
Dependence ρ	0 or 0.4
Innovation scale σ_t	Constant or increasing

- 1,000 datasets per scenario, 499 draws per method.
- Each dataset has 226 consecutive annual observations.
- Gaussian innovations and a central break at position 113.

8,000 datasets and 24,000 tests. Innovation scale is calibrated to 0.6182 °C. DWB uses $\ell = 6$. Master seed: 20260926. $x_t = (t - \bar{t})/10$; consecutive years do not reproduce the empirical gaps.

Ignoring dependence inflates false rejection



IID / INDEPENDENT WILD SIZE

26.5–32.2%

Under AR(0.4); nominal size is 5%

DWB SIZE

2.1–3.5%

Across scenarios: conservative here

95% Wilson intervals; panel scales differ. DWB power under AR(0.4): 74.6–78.5%. Higher apparent power does not compensate for inflated size. These designs do not establish empirical conservatism.

Later warming, qualified evidence of a break

A positive later fitted trend

+0.0887°C per decade; both interval constructions exclude zero.
The early slope's intervals include zero.

Break evidence depends on the bandwidth

Best-fitting date: 1851. The primary DWB test rejects at 5%;
bandwidths 12–15 do not. The date remains an uncertain model feature.

Dependence matters for inference

Independent methods over-reject in the AR scenarios.
DWB avoids that inflation, but remains an approximation.

The later slope averages 1851–2013, not recent decades. Missing years, one station and model assumptions limit broader or causal claims.

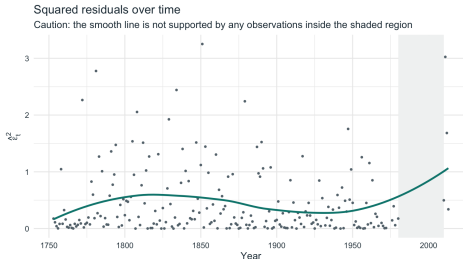
Self-reflection: strengths and limitations

- **What worked:** full sample, complete break reselection, three bootstrap methods and reproducible numerical checks.
- **What changed our conclusion:** residual dependence and bandwidth-sensitive break evidence.
- **Main limitations:** missing decades, one station, a single-break model and a heuristic bandwidth.
- **Next improvement:** evaluate interval coverage and bandwidth choices under a wider range of simulated dependence.

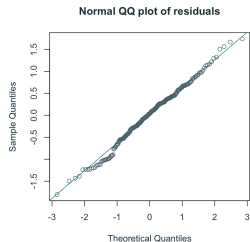
Two-minute self-reflection before Q&A. Interval coverage and wider bandwidth experiments are possible extensions; they were not performed.

Linear residual variance and marginal shape

Squared residuals over time



Normal QQ comparison



Local variance changes may escape a linear variance test.

The QQ centre is broadly linear, with tail departures.

The variance smooth across the gap lacks observed support. QQ assesses marginal shape; it cannot establish independence or validate the Gaussian reference model.

DW and BP use different references

DW Monte Carlo

Specify iid Gaussian errors and a fixed design X .

$$e^{(b)} = M_X z^{(b)}, \quad z^{(b)} \sim N(0, I_n).$$

Coefficients disappear under projection; the error scale cancels in DW.

Simulate separate lower and upper tails.

Gaussian simulation of BP is possible under a fully specified null.

It is not automatically valid for the broader homoskedasticity null. Squared residuals do not make simulation impossible; additional distributional assumptions are the issue.

BP asymptotic reference

Regress e_i^2 on $1, x_i$; compute $BP = nR^2$.

$$BP \stackrel{a}{\sim} \chi_1^2 \quad \text{under regularity conditions.}$$

Homoskedasticity alone does not specify the finite-sample distribution: error shape and dependence matter.

Generate each testing sample under the null

$$\tilde{e}_i = (e_{0i} - \bar{e}_0) \sqrt{n/(n-2)}, \quad y_i^* = \hat{y}_{0i} + e_i^*.$$

- **IID:** draw e_i^* independently with replacement from $\{\tilde{e}_j\}$.
- **Independent wild:** $e_i^* = \tilde{e}_i w_i$, independent equiprobable signs $w_i = \pm 1$.
- **DWB:** $e_i^* = \tilde{e}_i w_i$, with $w_1 \sim N(0, 1)$ and

$$a_i = e^{-(s_i - s_{i-1})/\ell}, \quad w_i = a_i w_{i-1} + \sqrt{1 - a_i^2} z_i, \quad z_i \stackrel{\text{iid}}{\sim} N(0, 1).$$

$$F_T^* = RSS_0^* - \min_k RSS_k^*, \quad p = \frac{1 + \sum_b 1\{F_{T,b}^* \geq F_T\}}{B + 1}.$$

Refit the null and all candidate alternatives in every draw; $B = 9,999$. Type-1 95% critical value, reject for $p \leq 0.05$. DWB preserves residual scale and calendar-distance multiplier dependence.

Break-test comparison and critical values

Procedure	95% critical value	p -value	At 5%
IID residual	2.643	0.0001	Reject
Independent wild	3.426	0.0001	Reject
Dependent wild ($\ell = 6$)	10.592	0.0331	Reject
Dependent wild ($\ell = 3$)	8.772	0.0206	Reject
Dependent wild ($\ell = 12$)	12.608	0.0541	Do not reject

Primary DWB: $p = 0.0331$; **twelve-year DWB:** $p = 0.0541$.

$F_T = 12.2106\text{ }^{\circ}\text{C}^2$; $B = 9,999$ per row. Critical values are in $^{\circ}\text{C}^2$. Independent methods have zero exceedances; $p = 0.0001$ is the simulation floor, not a zero probability.

Reported slope intervals and studentisation

Slope	Interval	Estimate	Lower	Upper	Contains 0
Pre-break	Percentile	-0.0525	-0.2601	0.0051	Yes
Pre-break	Percentile- t	-0.0525	-0.1804	0.0343	Yes
Post-break	Percentile	0.0887	0.0511	0.1551	No
Post-break	Percentile- t	0.0887	0.0318	0.1601	No

Percentile : $[q_{.025}(\hat{\theta}^*), q_{.975}(\hat{\theta}^*)]$,

$t^* = (\hat{\theta}^* - \hat{\theta})/s^*$, Percentile- t : $[\hat{\theta} - q_{.975}(t^*)s, \hat{\theta} - q_{.025}(t^*)s]$.

Nominal 95% intervals, °C per decade. DWB $\ell = 6$, 4,999 draws under the fitted alternative, full date reselection, type-7 quantiles; calendar HAC standard errors recomputed in every draw.

All eight simulation configurations

Quantity	ρ	Innovation scale	IID residual	Ind. wild	Dep. wild
Size	0.0	Constant	4.1	4.3	2.1
Power	0.0	Constant	100.0	100.0	99.0
Size	0.4	Constant	29.1	29.8	2.9
Power	0.4	Constant	97.7	97.5	78.5
Size	0.0	Increasing	8.8	5.4	2.6
Power	0.0	Increasing	99.9	99.7	98.5
Size	0.4	Increasing	32.2	26.5	3.5
Power	0.4	Increasing	97.7	96.6	74.6

Entries are rejection percentages; size uses $\delta = 0$, power uses $\delta = 0.14$. $M = 1,000$, $B = 499$, DWB $\ell = 6$. Dataset seed: $20260926 + 100000c + m$. Same datasets for all three methods.

DW definitions and calibration

$$d_{\text{obs}} = \frac{\sum_{i=2}^n (e_i - e_{i-1})^2}{\sum_i e_i^2}, \quad d_{\text{cal}} = \frac{\sum_{i: s_{i+1}-s_i=1} (e_{i+1} - e_i)^2}{\sum_i e_i^2}$$

Definition	Pairs	Statistic	Lower	Upper
Observation adjacency	225	1.4147	1.7549	2.2695
Calendar adjacency	220	1.3865	1.7106	2.2211

- Common 9,999 Gaussian draws, each projected off the design.
- Type-6 cutoffs are order statistics 250 and 9,750.
- Strict rejection outside cutoffs agrees with $p \leq 0.05$.

Both two-sided p-values are 0.0002. Exact rank size is conditional on the continuous iid Gaussian null and fixed design.

Testing and intervals use different models

Break test

Null mean $\hat{y}_0$ and residuals e_0 .

Scale residuals by $\sqrt{n/(n-2)}$.

Repeat the full search and count

$$F_T^* \geq F_T.$$

Slope intervals

Alternative mean $\hat{y}_1$ and residuals e_1 .

Scale residuals by $\sqrt{n/(n-3)}$.

Reselect the date and recompute slopes and standard errors.

$$J_i = [1, x_i, (x_i - \tau)_+, -\hat{\delta} 1(x_i > \tau)], \quad K_{ij} = \max\{1 - |s_i - s_j|/7, 0\}$$

Studentisation includes the local date direction and uses correction $n/(n-4)$. A rescaled indicator replaces the fourth column for numerical stability when the kink is nonzero. Weak identification can limit coverage.

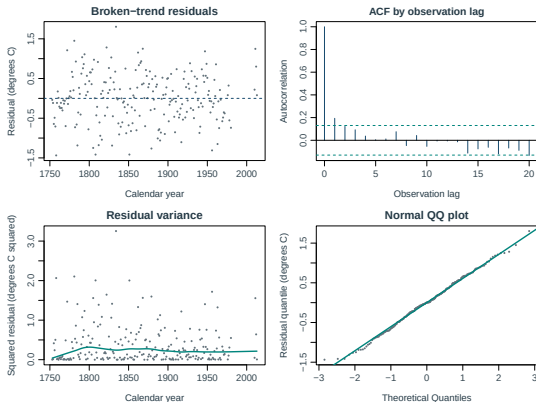
The main shape persists in reduced samples

Sample	n	Break year	Pre slope	Post slope
Full sample	226	1851	-0.0525	0.0887
Omit 1851	225	1853	-0.0464	0.0872
End in 1980	222	1850	-0.0477	0.0776

- Each reduced sample repeats the trimmed break search.
- These are descriptive checks without new bootstrap tests.
- Primary inference retains all 226 observations.

Slopes are $^{\circ}\text{C}$ per decade. Alternative-bootstrap date percentiles are 1793 and 1896, with 1.54% of selections at boundaries. These do not constitute a calibrated break-date confidence set.

Dependence remains after fitting the break



ACF(1):
0.290 to 0.194

DW:
1.415 to 1.613

RMSE:
0.657 to 0.614

R^2 :
0.105 to 0.217

Dependence remains.

Observation lags in the ACF. Selecting a date changes the design, so the linear-model DW critical values do not calibrate a test here.

AI assistance supported code debugging, writing clarity and presentation formatting; see the report for the project acknowledgement.